

An Overview of the Mechanical Keyboard

A mechanical keyboard is a computer input device that uses individual spring-loaded switches beneath each key to register keystrokes, offering tactile feedback and greater durability than a standard membrane keyboard. As seen in Figure 1, the mechanical keyboard consists of a rectangular case, rows of keycaps, and a cable or wireless receiver. Compared to a standard membrane keyboard, which typically costs between \$10 and \$30, a mechanical keyboard ranges from about \$50 for an entry-level board to over \$200 for an enthusiast-grade model. The purpose of this document is to inform readers who are unfamiliar with the mechanical keyboard about its features and how it works. The principal parts of the mechanical keyboard include the case, the switches, the keycaps, the connectivity options, and the form factors.



Figure 1: A full-size mechanical keyboard¹

The Case

The case is the outer shell of the keyboard that holds all of its internal components together. Cases are most commonly made from either ABS plastic or aluminum. A plastic case is lightweight and tends to produce a higher-pitched sound when you type, while an aluminum case is heavier and produces a deeper, more muted sound. The case also houses the printed circuit board (PCB), which is the internal board that each switch is mounted to. The quality of the case has a direct impact on how the keyboard feels and sounds during use.

Switches

The switches are the most important part of the mechanical keyboard. Each switch sits beneath a keycap and is mounted to the PCB. A switch consists of three parts: a housing, a stem, and a spring. When you press a key, the stem travels downward, compresses the spring, and completes an electrical circuit at the actuation point, which registers the keystroke on your computer.

Switches come in three types: linear, tactile, and clicky. As seen in Figure 2, a linear switch has a smooth, consistent keystroke with no feedback along the way. A tactile switch produces a noticeable bump midway through the press that lets you know the keystroke has been registered



Figure 2: Linear, tactile, and clicky switch types²

without bottoming out the key. A clicky switch produces the same bump as a tactile switch but also makes an audible click sound. Linear switches are popular for gaming because of their fast, uninterrupted keystroke. Tactile and clicky switches are popular for typing because the feedback helps you type more accurately. Common switch brands include Cherry MX, Gateron, and Kailh, and each brand offers its own versions of these three types.

Keycaps

The keycaps are the removable plastic caps that sit on top of each switch and are the part of the keyboard you actually touch when you type. Keycaps are most commonly made from one of two plastics: ABS or PBT. ABS keycaps are smooth and relatively thin, but they tend to develop a shiny, worn appearance over time from the oils on your fingers. PBT keycaps are thicker and have a slightly textured surface that resists shine and wear much better than ABS. As seen in Figure 3, because keycaps are removable, you can replace them with a different set to change the look and feel of your keyboard entirely.



Figure 3: ABS keycap (left) vs. PBT keycap (right)³

Connectivity

Mechanical keyboards connect to your computer either through a wired USB connection or wirelessly via Bluetooth. A wired connection uses a detachable USB-C or Micro-USB cable and provides a reliable, low-latency connection that does not require charging. A wireless connection uses Bluetooth or a 2.4 GHz USB receiver and gives you the freedom to use the keyboard without a cable on your desk. However, a wireless keyboard requires you to charge or replace the battery periodically. Most enthusiast-grade keyboards offer both wired and wireless modes so you can switch between them depending on your preference.

Form Factors

Mechanical keyboards come in several sizes, called form factors. As seen in Figure 4, the full-size layout has 104 keys and includes a number pad on the right side. The tenkeyless, or TKL, layout removes the number pad, bringing the key count down to 87. The 60% layout removes the number pad, the function row, and the arrow keys, leaving you with 61 keys. Smaller layouts are popular because they take up less space on your desk and allow you to keep your mouse closer to the keyboard. Most of the keys removed in smaller layouts can still be accessed by holding a function key and pressing another key in their place.



Figure 4: Full-size, tenkeyless, and 60% keyboard layouts side by side⁴

One Complete Operating Cycle

When you sit down to type, you press a keycap, which pushes the switch stem downward through the housing. The stem compresses the spring and reaches the actuation point, at which point the PCB registers an electrical signal and sends it to your computer through the USB cable or wireless receiver. Your computer receives the signal and displays the corresponding character on screen. When you release the key, the spring pushes the stem back up to its resting position, resetting the switch and preparing it for the next keystroke.

Conclusion

The case, switches, keycaps, connectivity, and form factor are all fundamental parts of the mechanical keyboard that work together to produce a typing experience that is more tactile, durable, and customizable than a standard membrane keyboard. The switch type you choose determines how each keystroke feels and sounds, the keycap material determines how the surface holds up over time, and the form factor determines how much space the keyboard takes up on your desk. Together, these parts make the mechanical keyboard one of the most customizable computer peripherals available.

Sources

¹ Image from <https://www.pcmag.com/reviews/best-mechanical-keyboards>

² Image from <https://atomoffice.com/blogs/buyers-guide-for-keyboards/keyboard-switch-types>

³ Image from <https://www.daskeyboard.com/blog/pbt-keycaps-vs-abs-keycaps-which-is-better/>

⁴ Image from <https://www.sportskeeda.com/gaming-tech/tkl-full-size-60-which-keyboard-form-factor-best-you>